APPLICATION NOTE

AN0052-EC NB-IoT Example Introduction_V1.1

EigenCOMM Wireless Microcontroller

Document Description

the purpose of present document is to introduce the respective purposes of the examples in the SDK

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Contents

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1. Brief

The present document introduce the respective purposes of the examples in the SDK. These examples are located in the SDK\ PLAT\project\ec616_0h00\apps. The directory structure is shown below. The following sections will introduce their functions and purposes one by one.

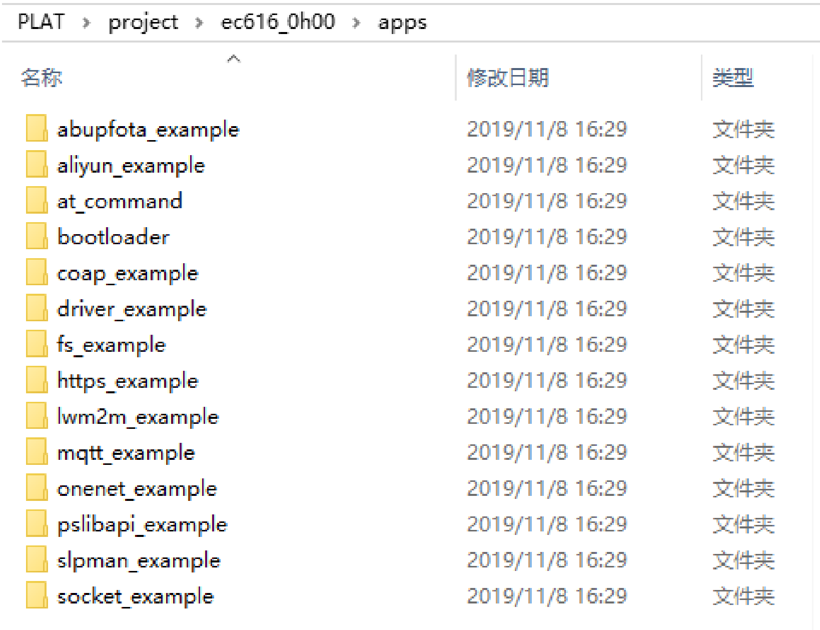

Figure 1. Directory structure

2. Example description

2.1 Abupfota example

ABUP is a company engaged in OTA business. This example uses the FOTA solution provided by ABUP, and completes the FOTA through the protocol define by ABUP that base on lwM2M.

The SDK port the ABUP's FOTA library at SDK\PLAT\middleware\thirdparty\abup. The example is implements the common interface "appInit" of SDK, which will call the FOTA entry in library of ABUP. Which is "abupfotaInit". It will start a thread to implement FOTA. The example process is shown below:

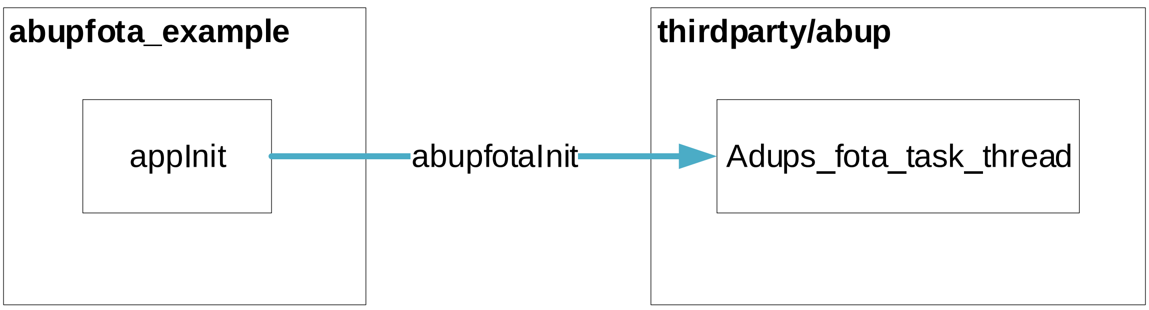

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Figure 2. abupfota example

2.2 Aliyun example

The SDK ported the Link Kit SDK provided by Alibaba Cloud. This SDK can connect device to Alibaba cloud IoT Platform through MQTT/CoAP/HTTP/HTTPS. This example create three client such as MQTT, CoAP, HTTP, by calling the API of the Link Kit SDK to connect with Alibaba Cloud.

The example process is shown below:

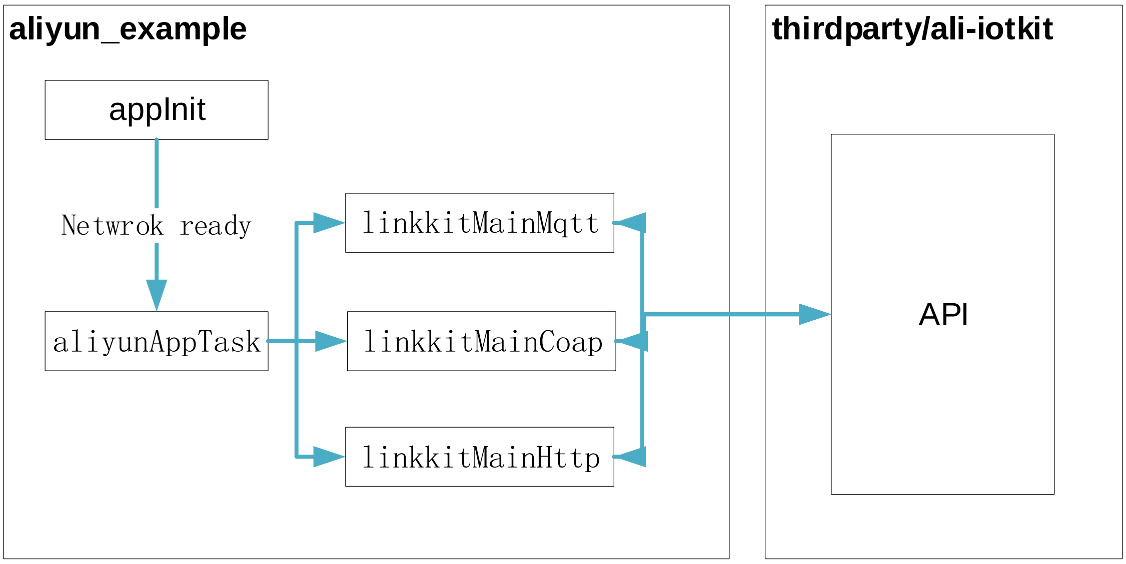

Figure 3. aliyun example

2.3 At command example

This is the default project of the SDK. It work with UART driver, receive AT command from UART and handle the AT command. For detailed description of AT command, please refer to the document "*EiGENCOMM EC616 AT Command Manual.docx*".

In addition, it handle the restore work for low power mode. These restore work are for onenet AT command, CTIOT AT command and CoAP AT command.

2.4 Bootloader example

The bootloader initializes essential clocks, drivers and load system image. The bootloader code will run integrity check with the image id and the hash value in the system image head if needed. The bootloader will then transfer to the system entry address.

2.5 Coap example

The SDK ported the CoAP library. Libcoap is open-source software. The example create a coap client through libcoap. The client will connect to CoAP server: coap.me, and send data to it. After receive the response from Sever, client will delete. With this example, user can master how to create client and send data using libcoap's

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API.

2.6 Ct example

The SDK integrates China Telecom’s SDK ctlwm2m. ct example use the API of that SDK. This example is suitable for scenarios that require low power consumption and need to send data to the China Telecomm platform periodically. The period can be one day or several hours. You can also use the key trigger to send data to the platform.

In this example, wakeuppad3 is the key to wakeup system and sending data.

Variable sendDataItv sets the period of sending data.

The ocean platform is the default platform the example connect.

If define Macro USE_CTWING_PLATFORM, example will connect to AEP platform.

The example mainly contains two files app.c and ct_api.c

Developers can modify app.c to suit their own applications. The ct_api.c integrates the API of ctlwm2m to realize the function of the lwm2m client.

2.7 Driver example

This example uses several demos to shows how to use various peripherals.

To ADC demo, the example keep sampling THERMAL, VBAT, AIO4 channels.

To USART, the example echo user’s input character one by one, USART work mode can be configured in RTE_Device.h(header file in example) to enable polling/interrupt/dma mode respectively.

To DMA demo, the example employ no-descriptor-fetch mode, also called as one-shot mode to perform memory to memory transfer.

To GPIO demo, the example toggle a LED on each button’s press.

To i2c demo, the example read the temperature of the temperature sensor on the board and output it to the log.

To SPI demo, the example leverage two instances of SPI in one board to communicate with each other, SPI work mode must be configured in RTE_Device.h(header file in example) to be DMA mode for work properly.

To TIMER demo, the example demos PWM feature of TIMER. It use two instance of TIMER. One is use to output PWM, and its frequency is configure as 1 KHz while another TIMER control the duty cycle. The duty cycle is updated in interval of five from zero to 100 periodically, the period is achieved with aid of another instance of timer whose interrupt is triggered every 500ms.

To WDT demo, example start to kick WDT for 5 times and then stop, the WDT is configured as WDT_InterruptResetMode and we won't clear WDT interrupt flag upon timeout, so a system reset shall follow a WDT interrupt, the time tick is achieved through TIMER's interrupt.

2.8 FS example

This is a simple example that updates a file named boot count every time example runs. The program can interrupted at any time without losing track of how many times it boot.

2.9 Https example

This example create an https client and connect to https server https://www.howsmyssl.com. It use GET method to get server's response and show it in log. The API used by the client is at SDK\PLAT\middleware\thirdparty\httpclient. It support http and https. User can follow this example to build their own http client.

2.10 Lwm2m example

The SDK ported the Wakaama. It is an implementation of the Open Mobile Alliance's LightWeight M2M protocol (LWM2M). The Wakaama provides an example client, and adds several standard objects for the client. Its server is "*leshan.eclipseprojects.io*", this URL may be invalid and user can update it. This example calls the API of wakaama to create this example client. In addition, user can operate the standard objects from server.

The example process is shown below:

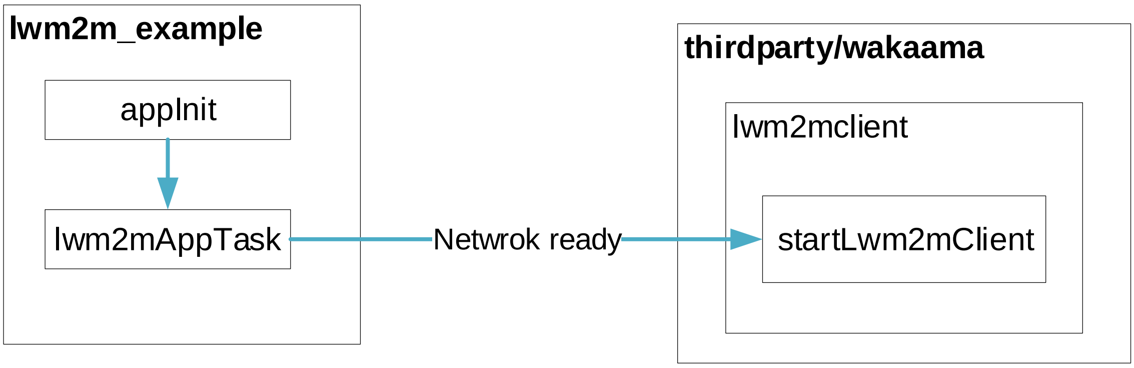

Figure 4. Lwm2m example

2.11 Mqtt example

The SDK ported the open-source code Eclipse Paho's sub-projects: MQTTPacket. The example create an mqtt client, which by default connect to OneNET's mqtt server. Then the example subscribe to server with topic "*test*", and then unsubscribe it. Then publish a message and delete the client. The example process is shown below:

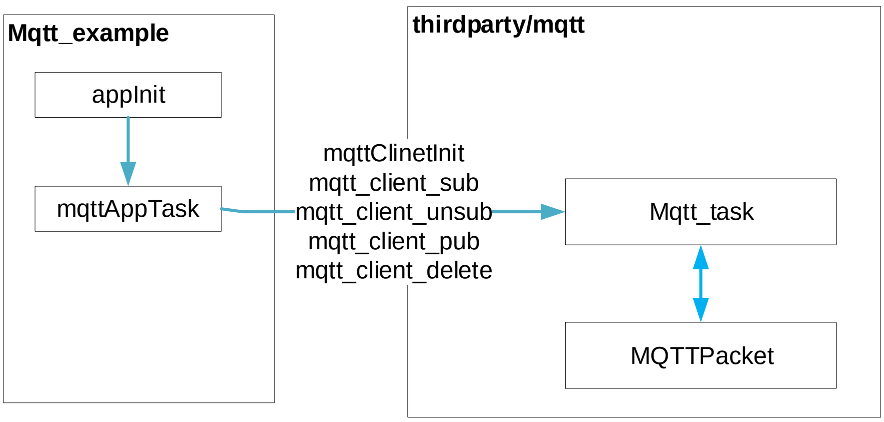

Figure 5. mqtt example

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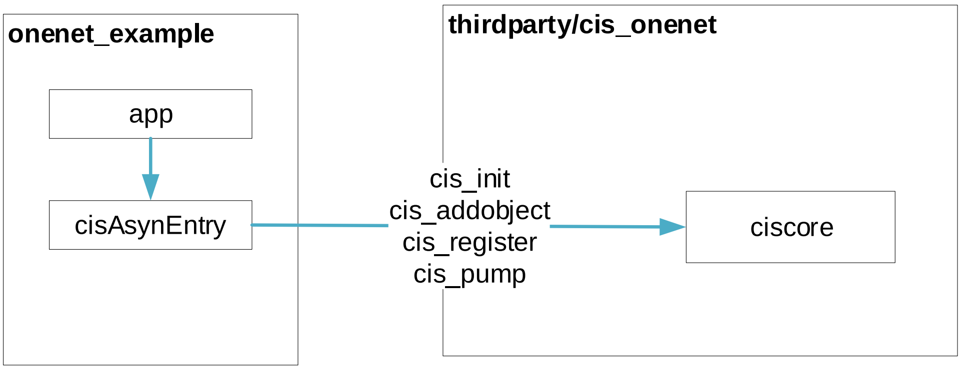
2.12 Onenet example

The SDK ported the NB_SDK2.3.0 provided by China Mobile. This SDK is base on wakaama. It can connect device to China Mobile's IoT Platform OneNET through LwM2M. This example create client by calling the API of the NB_SDK2.3.0 to connect with OneNET.

The example process is shown below:



2.13 Pslibapi example

This example show how to use the APIs of the protocol stack library. The APIs is in SDK\PLAT\middleware\eigencomm\psproxy\entry\src\pslibapi.c. Through these APIs, user can get the software version number, IMSI, TAC, Cell ID, SNR, coverage level, TAU information, APN information, system time, network information. And user can open SIM logical channel, generic SIM logical channel access, close SIM logical channel.

2.14 Slpman example

This example shows the use of sleep related APIs. The main APIs are as follows:

1. The application layer polls handles; the application layer can vote for allowed sleep levels.

2. Register backup callbacks for different sleep levels/Register restore callbacks for different sleep levels; Users can use these callbacks to do backup work before entering the corresponding sleep and restore work after ending the corresponding sleep.

3. Register a user deep sleep timer callback; this deep sleep timer can auto wakeup from hibernate when time is up.

2.15 Socket example

The SDK ported lwip – a Lightweight TCP/IP stack. The example realize a simple UDP server, a UDP client and a TCP client through the APIs of the lwip. They provide references for user.

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3. Version

Version	Date	Comments
Draft	2019-12-19	
V1.0	2019-12-19	
V1.1	2020-6-11	add ct example

4. About US

Shanghai EigenCOMM Technology Co.,Ltd is established in Feb 2017 in Zhangjiang Hi-Tech park, Shanghai China (www.eigencomm.com) . EigenCOMM focuses on cellular based IoT chipset and solution. With accumulation of experience over the years in algorithm, L1/L2/L3 protocol, SoC, RF , platform & application software as well as the communication architecture and system, team makes its way to IoT industry.

The ultra-low power NB-IoT chipset EC616(S) has already enter mass production stage. The 4G Cat.1 chipset EC618 is in engineering samples stage, 5G is also in the R&D stage.

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